JNU MCA - 2018

1. Find the solution of $\lim_{0\to 0} \frac{\sin\ \theta}{\theta}$
   a. 0
   b. 1
   c. θ
   d. $1/_2$
2. To remove a relation in SQL database we use _____ command.
   a. Remove
   b. Purge
   c. Deleted
   d. Drop table
3. $\int \log\ x\, \mathrm{d}x$
   a. $x\log(x / e)$
   b. $x\log(e / x)$
   c. $\mathrm{x}\log\mathrm{x}$
   d. $1/x$
4. Which is not leap year?
   a. 700
   b. 800
   c. 1200
   d. 2000
5. If 4 coins are thrown together then probability that at least one head occurs
   a. $5/_{16}$
   b. $3/_8$

c. $^{1}/_{16}$
d. $^{15}/_{16}$

6. Which is not irrational number
   a. π
   b. $\sqrt{2}$
   c. $\sqrt{3}$
   d. $\sqrt{4}$
7. Convergence sequence has limit
   a. Only one limit
   b. Only two limit
   c. Only three limit
   d. None
8. The operation through NOT GATE is also
   a. Converting
   b. Inverting
   c. Reverting
   d. Reversing

9. A is 3 year older than C & 3 year younger than B. If B and D both are twins, than how many years C
   a. 8
   b. 6
   c. 7
   d. None
10. **Statement & Conclusion**
    - **Statement:**

      All mango is golden colour.

      No golden colour things are cheap

    - **Conclusion:**
      I. All mangoes are cheap.
      II. Golden coloured mangoes are not cheap

a. Only conclusion **I** is true.
b. Only conclusion **I I** is true.
c. Either conclusion **I** or **I I** true.
d. Neither conclusion **I** nor **I I** is true

11. Find the Eigen vector corresponding to Eigne value $\lambda = 2$ for the metrix $A = \begin{bmatrix} 5 & 3 \\ 2 & 4 \end{bmatrix}$
    a. $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$
    b. $\begin{bmatrix} 3 \\ -2 \end{bmatrix}$
    c. $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
    d. $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
12. If u=$\tan^{-1}\left[\frac{x^3+y^3}{x+y}\right]$ then find value of $x\frac{\partial a}{\partial x} + y\frac{\partial u}{\partial y}$?
    a. Sin u
    b. Sin 2u
    c. Cos u
    d. Cos 2u

13. If integer 2 bytes of storage, then what is the maximum value of unsigned integer.
    a. $2^{16} - 1$
    b. $2^{15} - 1$
    c. 2^{15}
    d. 2^{16}
14. The given equation of the circle $x^2 + y^2 + 2x + 2ky + 6 = 0$ & $x^2 + y^2 + 2ky + k = 0$ intersect each other orthogonally, then find the value of k
    a. $-2, -{}^{3}/_{2}$
    b. $2, -{}^{3}/_{2}$
    c. $2, {}^{3}/_{2}$
    d. $-2, {}^{3}/_{2}$
15. If $f(x) = x^2 + \frac{x^2}{(1+x)^2} + \dots\dots\dots.\frac{x^2}{(1+x)^n} + \dots..$ then at x=0
    a. $\lim_{x\to 0}$ f(x)does not exists.
    b. $\lim_{x\to 0}$ f(x) exist but f(x)is not continuous

c. f(x) is continuous

d. None of these

16. The last term of the sequence 8, 6, 9, 23, 87,__?

a. 128

b. 224

c. 324

d. 429

17. If M= a cos θ + b sin θ and n= a cos θ − b sin θ then find the value of $m^2 + n^2 = ?$

a. $(ab)^2$

b. $(a+b)^2$

c. $a^2 + b^2$

d. 1

18. The two number's are in the ratio 3:5 if 9 is subtracted from each of the 2 no's , then the new number's are in the ratio 12:23, then smaller number is

a. 52

b. 33

c. 41

d. 27

19. The angle of elevation of sum when shadow of tree is equal to 3 times of the height of the tree is

a. 30^0

b. 60^0

c. 45^0

d. 90^0

20. If the equation $ax^2 + bx + c = 0$ has repeated roots. Then find the relation between coefficients.

a. $\left(\frac{b}{2}\right)^2 = c$

b. $\left(\frac{b}{2}\right)^2 = \mathrm{a}c$

c. $\frac{a^2}{2} = bc$

d. $\left(\frac{b}{2}\right)^2 = \mathrm{b}c$

21. A pair of dice (faces marked up to 1 to 6) is thrown then what is the chance of obtaining 6 on one of them give that the sum of the no's on the 2 dice is 8

    a. $\frac{35}{36}$
    b. $\frac{5}{36}$
    c. $\frac{2}{5}$
    d. $\frac{4}{5}$

22. The equation $x^3 + x^2 + x + 1$ when divided by $x^2 + x + 1$ leaves the remainder

    a. X
    b. X+1
    c. 0
    d. 1

23. $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos}} \mathrm{d}x$

    a. $\frac{\pi}{2}$
    b. $\frac{\pi}{4}$
    c. π
    d. 0

24. A rational database consist of a collection of

    a. Table
    b. Fields
    c. Records
    d. Keys

25. Consider the line passing (1,2) & (4,8) then the gradient is equal to

    a. $\frac{1}{2}$
    b. 2
    c. $\frac{-1}{2}$

d. -2

26. What is the full form of SQL
    a. Standard Query Language
    b. Sequential Query Language
    c. Structured Query Language
    d. Server side Query Language

27. Value of $\sqrt{\frac{1}{2}}$
    a. $\sqrt{\pi}$
    b. π
    c. $\sqrt{\frac{\pi}{2}}$
    d. $\frac{\pi}{2}$

28. If Sin $\theta + Cos\ \theta$ =1 then find the value of Sin 2θ
    a. 0
    b. 1
    c. 2
    d. None

29. Matrix $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 1 & 4 \end{bmatrix}$ satisfies which of the following properties
    a. Orthogonal
    b. Invertible
    c. Symmetric
    d. Singular

30. nth derivative of the term sin(ax+b)
    a. a sin(ax+b+nπ)
    b. a^n sin(ax+b+ $\frac{n\pi}{4}$)
    c. a^n sin(ax+b+ $\frac{n\pi}{2}$)
    d. a sin(ax+b+2nπ)

31. The sequence $\frac{(-1)^n}{n}$

a. Divergent
b. Convergent
c. Bounded
d. Unbounded

32. Sum of product can be implemented by the group

a. OR Gate
b. AND Gate
c. NOT Gate
d. XOR Gate

33. Which of the following is not true about Zero

a. Even
b. Positive
c. Additive identity
d. Additive inverse & zero

34. In u= tan $\left(\frac{\pi}{4}+\frac{\theta}{2}\right)$ then find the value of tan $h\frac{\theta}{2}$

a. Tan θ
b. Tan $\frac{\theta}{2}$
c. Tan $\frac{\theta}{4}$
d. tan 2θ

35. How many different committees of 3 people in a class room of 10 students?

a. 120
b. 24
c. 20
d. 12

36. Eigen value of the matrix$\begin{bmatrix} 4 & 1 \\ -1 & 2 \end{bmatrix}$

a. 3,3
b. 4,2

c. 1,-1

d. -3,3

37. If Y=X^{x^x} then $x\frac{dy}{dx}$ =?

a. $\frac{y^2}{1-y\log\, x}$

b. $\frac{y^2}{y\log x-1}$

c. $\frac{y}{1-y\mathrm{l}\quad x}$

d. $\frac{y^2}{1+y\log\, x}$

38. Father of C Language

a. James Gosline

b. B. Jame stroutup

c. Dennis Ritchie

d. Dr. E.F. Codd

39. If a plane passes through the point (2,3,-1) & it is right angle to OP where O is the origin then find the equation of the plane which is perpendicular to OP & passes through P.

a. 2x+3y-z=14

b. 2x+3y+z=14

c. 2x-3y-z=14

d. 2x-3y+z=14

40. Number of solution in $\begin{bmatrix}1 & 0 & 5\\ 0 & 1 & 6\\ 1 & 0 & 5\end{bmatrix}\mathrm{x}=\begin{bmatrix}1\\ 2\\ 3\end{bmatrix}$

a. 0

b. 1

c. Infinite

d. None of these

41. Find the greatest common division of $3^{13}*5^{17}$ & $2^{12}*3^{5}$is

a. 3^0

b. 3^1
c. 3^2
d. 3^5

42. The worst case running time to quick sort is
    a. 0(log n)
    b. 0(n log n)
    c. 0(n)
    d. 0(n^2)
43. Find the value of determinant $\begin{vmatrix} 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{vmatrix}$
    a. 3
    b. 2
    c. 1
    d. 0
44. If 75% students in a class study probability & 30% of the students study statics & the students who study both probability & statics is 20% , then find the probability of students who either study statics & probability is
    a. 75%
    b. 85%
    c. 95%
    d. None
45. I : If enrollment of class A is higher than B

    II : If enrollment of class C is lower than B

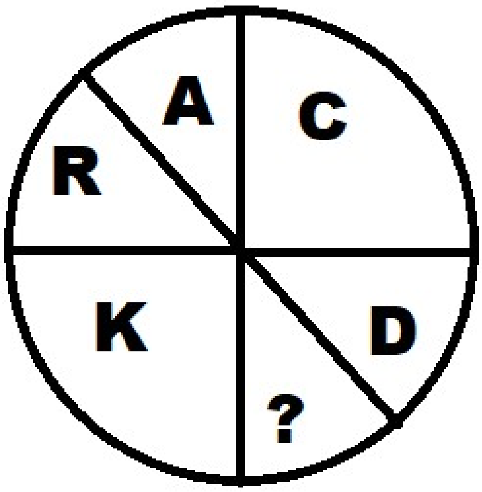
    III : Enrollment of A is lower than C

    If I and II is true then III is

    a. True
    b. Uncertain
    c. False

d. None of these

46. Find the letter which will come in the place of question mark?



a. F
b. G
c. H
d. I

47. Which is the smallest integer $\left[\frac{1+i}{1-i}\right]^{n}$=1 where i^2=-1

a. 2
b. 3
c. 4
d. 5

48. Let a & b are the 2 vectors given by $\vec{a} = 2\hat{i} - 3\hat{J} - \hat{k}$ & $\vec{b} = \hat{i} + 4\hat{J} - 2\hat{k}$ then the cross product $\vec{a} * \vec{b}$

a. $10\hat{i} - 3\hat{J} + 11\hat{k}$
b. $10\hat{i} + 3\hat{J} - 11\hat{k}$
c. $10\hat{i} - 3\hat{J} - 11\hat{k}$
d. $10\hat{i} + 3\hat{J} + 11\hat{k}$

49. The convergence of the following method is sensitive to the starting value is,

a. False Position Method
b. Gauss Siedal Method
c. Newton Ralphson
d. All of these

50. X(x - y)dy +y^2dx = 0, the solution of differential equation is

a. Y=c$e^{y/x}$
b. Y=c$e^{x/y}$
c. Y=cx$e^{y/x}$
d. x=cy$e^{y/x}$

51. Find the significant no. of digits in 204.020050

a. 5
b. 6
c. 8
d. 9

52. What is the last group of the series :
JAK KBL LCM MDL ___

a. OPS
b. QFN
c. NEO
d. PAQ

53. What is the postfix expression of infix expression A + B * C

a. AB + c*
b. ABC *+
c. +AB *C
d. ABC +*

54. Which of the following applies to the bisection method which is required for finding roots:

a. Converging to all other function.
b. Guaranteed work to all other function.

c. It is faster than Newton Ralphson’s method.
d. Require that there to be non-determination positive value.

55. If for real value of a cos θ= x+1/x then
a. θ is acute angle
b. θ is obtuse angle
c. θ is right angle
d. No value of θ is possible

56. Here some words translated for artificial language

Granamelke means big tree

Pinimelke means title tree

Melkehoon means tree house

Which word would mean big house

a. Granahoon
b. Pinihoon
c. Granamelke
d. Melkepini

57. The pointer pointing to “NOTHING” is called
a. VOID pointer
b. DANGLING pointer
c. NULL pointer
d. WILD pointer

58. Which of the following join condition contains the equality operator
a. Equijoins
b. Natural join
c. Cartesian join
d. Left join

59. The degree of Multiprogramming is
a. No. Of process in ready queue

b. No. Process executed per unit time
c. No. Of process in I/O
d. No. Of process in memory

60. How many time you write digit 3 between 1 to 100
a. 11
b. 18
c. 20
d. 21

61. If 100 cats kill 100 mice in 100 days then 4 cats kill 4 mice in how many days
a. 10
b. 4
c. 40
d. 100

62. X=101100, Y=1000011, then find X – Y using 2's compliment
a. 1010001
b. 1100101
c. 100110
d. 10001

63. Cost of pineapple is ₹ 7 & cost of watermelon is ₹ 5 & X spent 38 on these fruits, then how many pineapple were bought by X ?
a. 2
b. 3
c. 4
d. Data inadequate

64. $(256)^{0.16} * (256)^{0.09}$
a. 4
b. 16
c. 64
d. 128

65. Find the focus of the parabola y=-2$(x+4)^2$-1

a. (2,8/11)
b. (-2,-8/11)
c. (4,9/8)
d. (-4,-9/8)

66. The no. of attributes in a relation is called as :
a. Cardinality
b. Degree
c. Tuple
d. Entity

67. If x = -1 is a root of the equation $x^3 - 4x^2 - 89x - 84$ then other root is
a. 12
b. 7
c. 3
d. 6

68. If α, β are the roots of the equation $x^2 + 7x + 8 = 0$ then what is the value of $\alpha^2 + \beta^2 + \alpha.\beta$
a. 7
b. 17
c. 41
d. 49

69. The most significant bit with arithmetic addition is called
a. Overflow
b. Carry
c. Output
d. Zero

70. If element ABCD are inserted into the stack then what the sequence to removal
a. ABCD
b. DCBA
c. DCAB

d. ACBD

71. Chose the odd one out

a. Lotus
b. Rose
c. Bud
d. Tulip

72. $\left[\frac{1}{\log_4 120}+\frac{1}{\log_5 120}+\frac{1}{\log_6 120}+\cdots\dots\dots\dots\dots\right]$

a. 0
b. 1
c. 5
d. 120

73. If θ be the angle between $\vec{a}$ & $\vec{b}$ then what is the value of sin $\theta/2$

a. $\frac{1}{2}\left|\vec{a}-\vec{b}\right|$
b. $\frac{1}{2}\left|\vec{a}+\vec{b}\right|$
c. $\left|\vec{a}+\vec{b}\right|$
d. $\left|\vec{a}-\vec{b}\right|$

74. let R be a relation on N = {1,2,3,.........} defined by R = { (x , y) : x +2y = 10 } x ∈ N then y ∈ N then relation R is

a. Reflexive
b. Symmetric
c. Ant symmetric
d. Transitive

75. What is the next successive value of x_1 for the function f(x) = $x^2 - 2$ where x_0 =1, by using Newton Raphson's method

a. 1.315
b. 1.5
c. 2

d. 2.5

76. What is process

a. Program in secondary memory
b. Program in execution
c. Program in high level language kept in memory disk
d. Program in main memory

77. If A and B are two non empty sets having 5 elements in common, then number of elements common in A x B and B x A is

a. 2^5
b. 5^2
c. 10
d. 9

78. Consider the universal set U = {1,2,3,4,5} & set A = {1,2} & set B = {1,2,3,4}. Then set $A \cup \bar{B}$

a. {1,2,3,4,5}
b. {1,3,5}
c. {2,3,4}
d. {1,5}

79. If has been established that

P : Einstein was

Q : Although a great scientist

R : Weak in arithmetic

S : Right from his school days

The correct sequence is

a. SRPQ
b. QPRS
c. QPSR
d. RPQS

80. Runtime mapping from virtual to physical memory is done by :
   a. Memory Management
   b. CPU
   c. PCI
   d. None

81. Match the pair :

A. Newton Raphson	1. Integration
B. Runge's Kutta	2.Differential equation
C. Gauss Seidal	3.For finding root
D. Simple Rule	4.Solution of linear equation

Which of the following is correct :

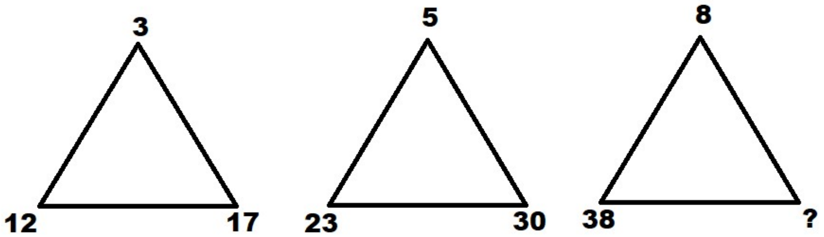
   a. A B C D 2 3 4 1
   b. A B C D 3 4 1 2
   c. A B C D 4 1 2 3
   d. A B C D 1 2 3 4

82. Find the missing term :

53	**J**	**49**
82	**X**	**37**
36	**L**	**15**
14	**?**	**98**

   a. B
   b. C
   c. E
   d. D

83. Find the missing term :



a. 45
b. 46
c. 47
d. 44

84. Given operation P ⊗ Q shows

P	Q	P⊗Q
T	T	T
T	F	F
F	T	F
F	F	F

a. Conjunction
b. Disjunction
c. Negation
d. Injection

85. If x=r cos θ and y = r sin θ then find value the of $\left(\frac{\partial r}{\partial x}\right)^2 + \left(\frac{\partial y}{\partial x}\right)^2$

a. 0
b. 1
c. 2
d. 3

86. In a binary search tree which of the following traversal would print the list in ascending order

a. Level order traversal
b. Pre order traversal
c. Post order traversal
d. In order traversal

87. The curved surface area of hemisphere is 175 cm^2. What is the radius of hemisphere
a. 3.25
b. 3.48
c. 3.38
d. 5.28

88. Choose the appropriate deadlock avoidance algorithm
a. Round Robin algorithm
b. Bankers algorithm
c. Elevator algorithm
d. Karn's algorithm

89. A unit vector perpendicular to the plane of $2\hat{i} - 6\text{j} - 3\hat{k}$ & $4\hat{i} + 3\hat{j} - \hat{j}$ is
a. $\frac{3}{7}\hat{i} - \frac{2}{7}\hat{j} + \frac{6}{7}\hat{k}$
b. $\frac{3}{7}\hat{i} + \frac{2}{7}\hat{j} + \frac{5}{7}\hat{k}$
c. $\frac{3}{7}\hat{i} - \frac{4}{7}\hat{j} + \frac{6}{7}\hat{k}$
d. $-\frac{3}{7}\hat{i} - \frac{4}{7}\hat{j} - \frac{6}{7}\hat{k}$

90. The word "break" is not used in
a. If else
b. do while
c. for
d. while

91. If the ages of the parallelepiped are the vectors $\hat{i} + \hat{j} - \hat{k}$, $2i + 3j$, $\hat{i} - 3\hat{k}$ then what is value of parallelepiped

a. 1
b. 4
c. 8
d. 6

92. Which of the following commutative
a. Subtraction of real number
b. Division of positive number
c. Multiplication of n x n matrix
d. Addition of n x m matrix

93. How many digits are used in numbering a book of 366 pages ?
a. 1093
b. 90
c. 990
d. 1305

94. What is the minimum of 2 input NAND Gate is used to perform the function of 2 input OR Gate ?
a. One
b. Two
c. Three
d. Four

95. Link list contains two fields one field is used to store the data, for what the second field is used for:
a. Pointer of character
b. Pointer to integer
c. Pointer to node
d. None

96. The surface area of the solid generated by revolution $x^2 + (y - b)^2 = a^2$, b $\geq$ a rotated about x – axis is
a. π^2ab
b. $2\pi^2$ab

c. $3\pi^2$ab

d. $4\pi^2$ab

97. Time quantum is used in

a. Shortest job first

b. Round Robin Scheduling

c. Priority Scheduling

d. None

98. Find the value of scalar m, if vector $2\hat{i} + \mathrm{m}\hat{\mathrm{j}} + \hat{k}$ & $4\hat{i} - 2\hat{j} - \hat{k}$ are perpendicular

a. 2

b. 1

c. 3

d. 5

99. Which one of the following is not a reserved key for C ?

a. Auto

b. Case

c. Main

d. Default

100. What argument can be made in 3 identical blue balls & 2 identical green balls in a straight line

a. 120

b. 24

c. 20

d. 10

Answer key

1	B	26	C	51	D	76	B
2	D	27	A	52	C	77	B
3	A	28	A	53	B	78	D
4	A	29	D	54	B	79	B
5	D	30	C	55	D	80	A
6	D	31	C	56	A	81	A
7	A	32	A	57	C	82	B
8	B	33	B	58	A	83	C
9	B	34	B	59	D	84	A
10	B	35	A	60	C	85	B
11	C	36	A	61	D	86	D
12	B	37	A	62	D	87	D
13	A	38	C	63	C	88	B
14	B	39	A	64	A	89	A
15	C	40	A	65	D	90	A
16	D	41	D	66	B	91	B
17	Wrong	42	D	67	A	92	D
18	B	43	B	68	C	93	C
19	Wrong	44	B	69	B	94	C
20	B	45	C	70	B	95	C
21	C	46	B	71	C	96	D
22	D	47	C	72	B	97	B
23	B	48	D	73	C	98	C
24	A	49	C	74	B	99	C
25	B	50	A	75	B	100	d